
A MULTIVARIATE LINEAR MODEL FOR AGRICULTURAL DATA OF IOWA FOR CORN, SOYBEANS, AND WHEAT

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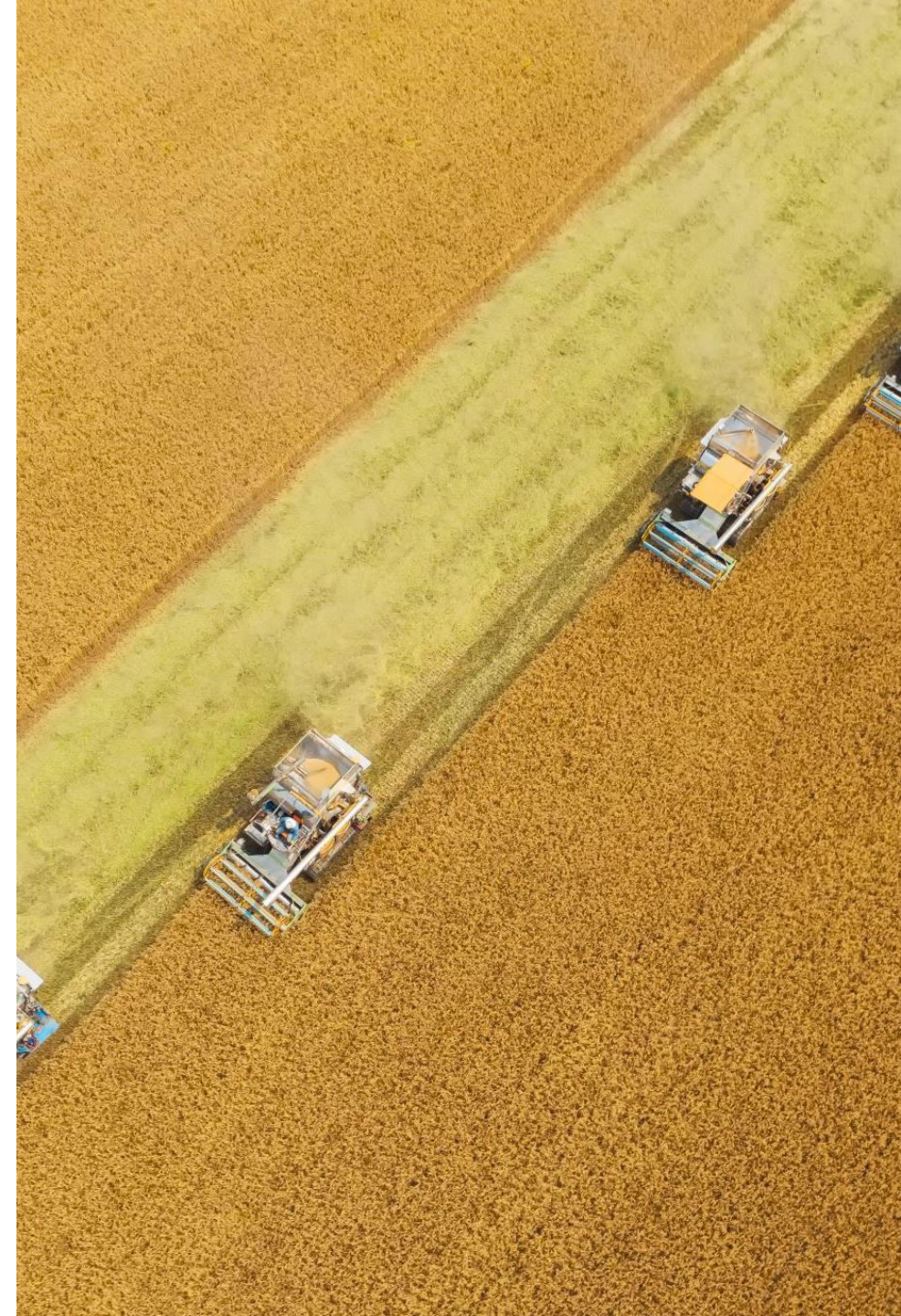
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INTRODUCTION/BACKGROUND

- A multivariate linear model is statistical method of prediction that is used to conduct analysis of covariance between **multiple dependent variables(outcomes)** concurrently as a **linear combination of the predictor variables**.
 - Multiple Independent and Dependent Variables
- Models a vector of responses capturing the relationships between outcomes of the predictors with the goal of understanding how factors impact multiple outcomes simultaneously.

INTRO & BACKGROUND CONT'D

- A Multivariate Linear Model follows:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \text{ or } \mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$$

- $\mathbf{Y}$ is an $n \times 1$ vector of the dependent variables
- $\mathbf{X}$ is the $n \times (p + 1)$ matrix that represents the design matrix of the independent variables
- $\mathbf{B}(\text{beta})$ is a $p + 1 \times 1$ vector and it is the coefficient or parameters to be estimated
- $\mathbf{E}(\text{epsilon})$ is an $n \times 1$ vector and it is the error terms

Purpose: Allows the compact solving of linear regressions with multiple predictors

INTRO CONT'D

SOLVING FOR PREDICTORS

- Ordinary Least Squares (OLS)

- Top Estimate Coefficients

- $\mathbf{B} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$

- $\mathbf{X}$: Design Matrix; input data

1	X_{11}	X_{12}	X_{13}	X_{14}
1	X_{21}	X_{22}	X_{23}	X_{24}
1	X_{31}	X_{32}	X_{33}	X_{34}
1	X_{41}	X_{42}	X_{43}	X_{44}
1	X_{51}	X_{52}	X_{53}	X_{54}

- $\mathbf{X}^T$: Transpose of the Design Matrix

1	1	1	1	1
X_{11}	X_{21}	X_{31}	X_{41}	X_{51}
X_{12}	X_{22}	X_{32}	X_{42}	X_{52}
X_{13}	X_{23}	X_{33}	X_{43}	X_{53}
X_{14}	X_{24}	X_{34}	X_{44}	X_{54}

- $(\mathbf{X}^T \mathbf{X})$ Multiplication of both matrices which only exist if there is no multicollinearity (full column rank)
 - $(\mathbf{X}^T \mathbf{X})^{-1}$ The inverse of that matrix

INTRODUCTION/BACKGROUND CONT'D

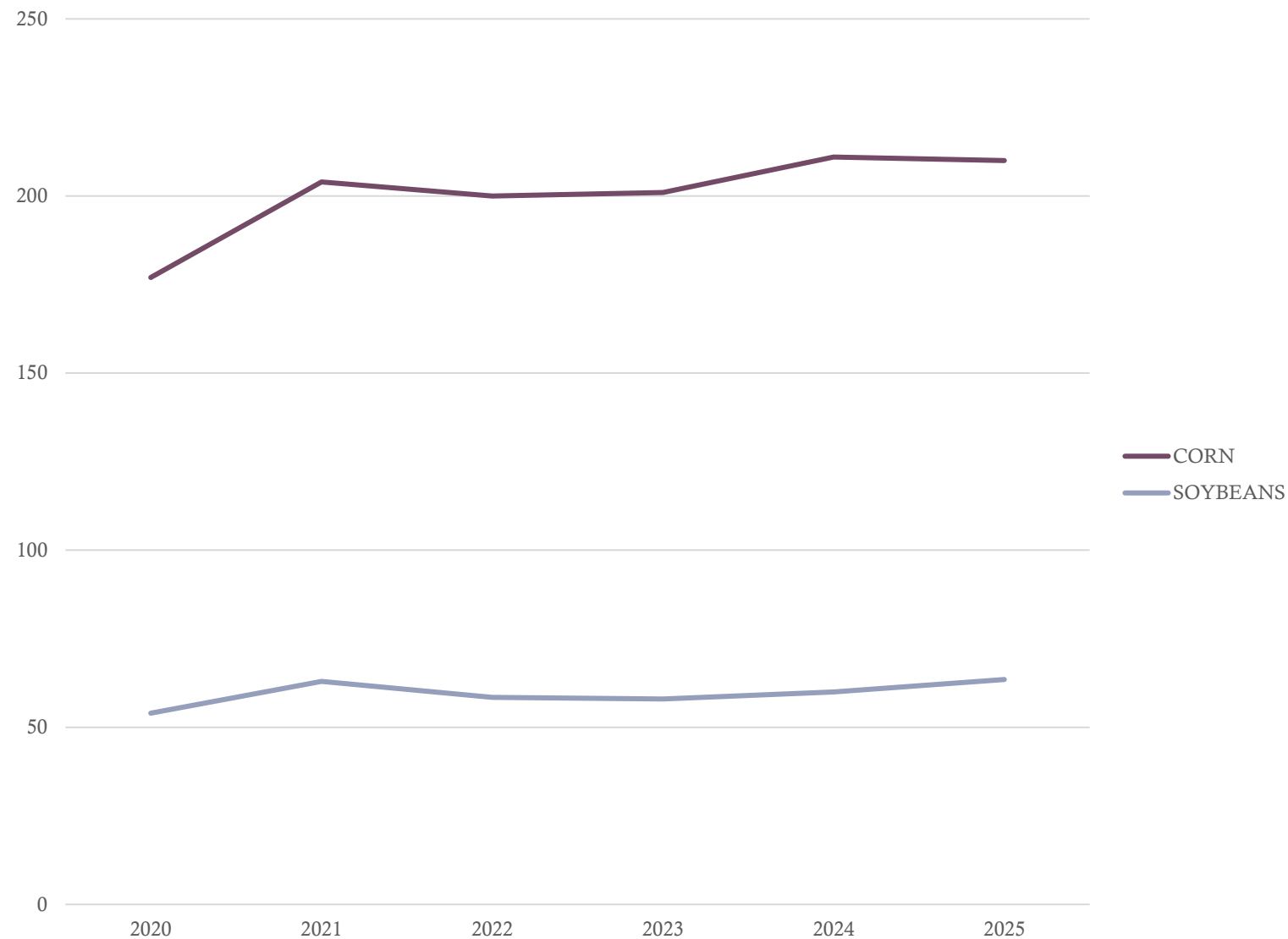
ABSTRACT: APPLICATION OF THE CONCEPT

- This project investigates the joint relationship between Corn, Soybeans, and wheat yield in Iowa.
 - Particularly climatic variables such as precipitation, temperature and anthropogenic factors like fertilizer(Nitrogen) use.
 - To fill the gaps that already exist in the univariate models.
 - We took data from the USDA NASS from 2020 to 2025, with a focus on soil quality, considering the covariance between corn, soybeans, and wheat yields, and other means to address missing data.
 - The model follows the form $\mathbf{Y}=\mathbf{XB}+\mathbf{E}$
 - Y is the multi-crop yield matrix,
 - X as the matrix for environmental predictors
 - E the error terms associated.
-

IOWA AS AN AGRICULTURAL POWERHOUSE AND THE “*POWERFUL TAG TEAM*” OF CORN AND SOYBEAN

- The “dynamic duo” started off as horse forage crop, and with time became the predominant grain crop because of various historical and biological factors.
 - **Agronomical Synergy:** Soybeans are classified as *legumes* with the **natural ability to improve nitrogen in the soil**, due to this, corns will require less nitrogen fertilization, which reduces costs for farmers.
 - **The rotation** of these two crops on a farm provides **pest and disease control**, resulting in better health for crops and higher yields.
 - **Residue Management:** the adaptability of soybeans, they can thrive regardless of the heavy corn residue
 - They also **use the same** machinery.

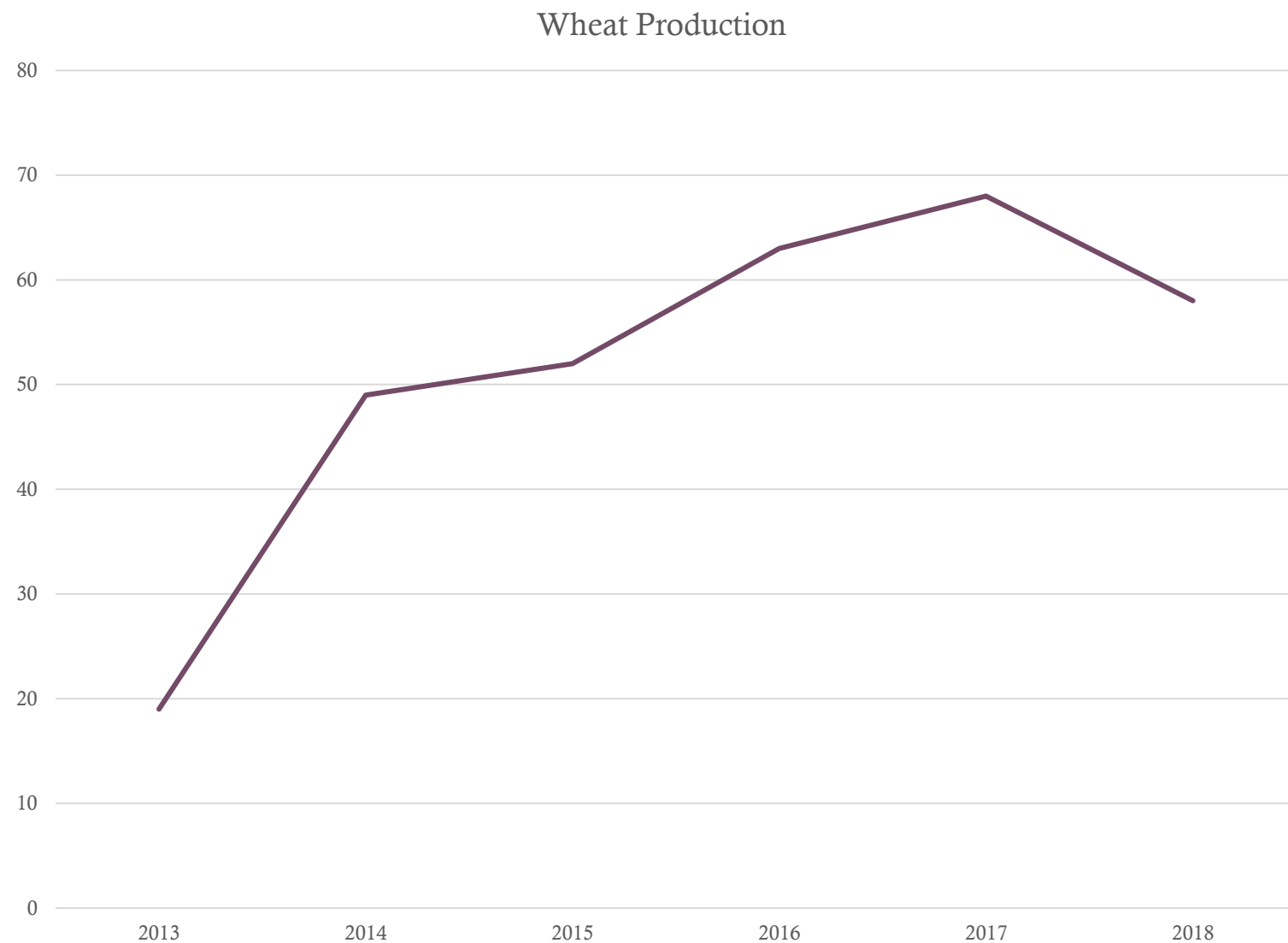
CORN & SOYBEAN CONT'D YIELD FROM 2020-2025 (SIX-YEAR PERIOD)



WHEAT PRODUCTION IN IOWA

- In search of data, we found that the production of Wheat has decreased and is practically non-existent in Iowa. The latest data on USDA NASS was in 2018.
 - Farmers in Iowa were surveyed in the *Iowa Farm and Rural life Poll* in 2025 (Gordon Arbuckle Jr., 2025)
 - The survey revealed the financial challenges that farmers face that causes this crop production dynamic (Corn & Soybeans)
 - **“Rising input cost and profit margins”**, input costs like seed and chemicals tends to increase when “profit margins are higher” 88% of them agreed that these profit margins on corn and soybeans are lost to land rent and input costs, therefore “underscoring persistent financial pressures in row crop production.”
 - **Uncertainty and Risk in the economy**, 85% of famers agreed or strongly agreed that “the boom-bust cycles of commodity production are hard on farmers”
 - 69% feel sometimes like **they have minimal control over profitability**,
 - 67% agreed **that heavy reliance on the financial risk of the heavy reliance on corn and soybeans**,
 - 60% felt **“overly dependent on purchased inputs.”**
- To conduct our analysis and model the data regardless of the lack of recent data, we used the most recent six-year time period which is from 2013-2018

SIX YEAR TIME SERIES OF WHEAT YIELDS IN IOWA 2013-2018



NOTE

- It is important to note that, there is a multitude of missing data, we try our best to conduct our analysis, data modeling and prediction with the best information available.

VARIABLE IN OUR MODEL



X_1 (PRECIPITATION): TOTAL
RAINFALL CRITICAL
WINDOW (APRIL-AUGUST)



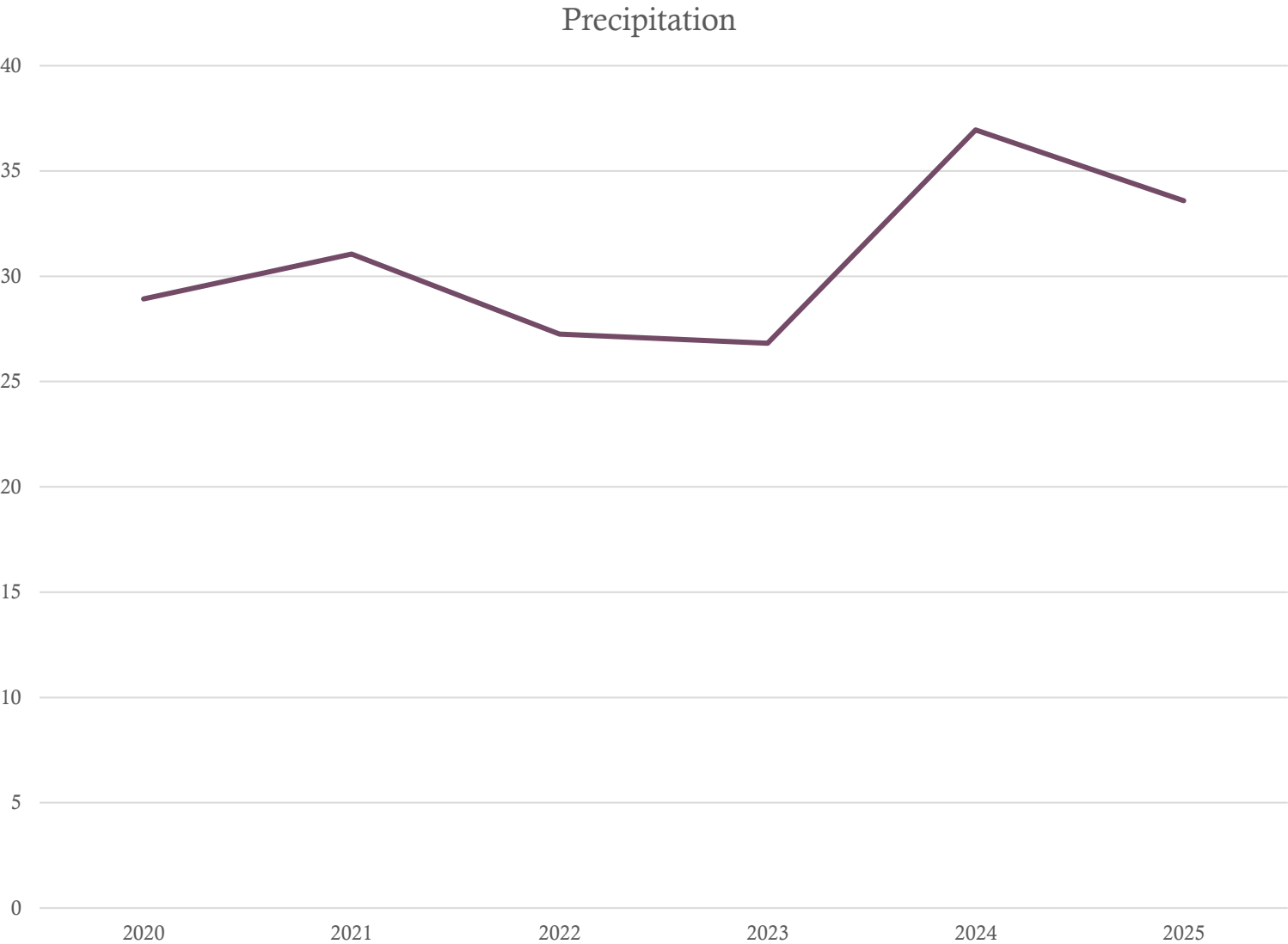
X_2 (TEMPERATURE): AVERAGE
DAYTIME TEMPERATURE OF
“GROWING DEGREE DAYS”



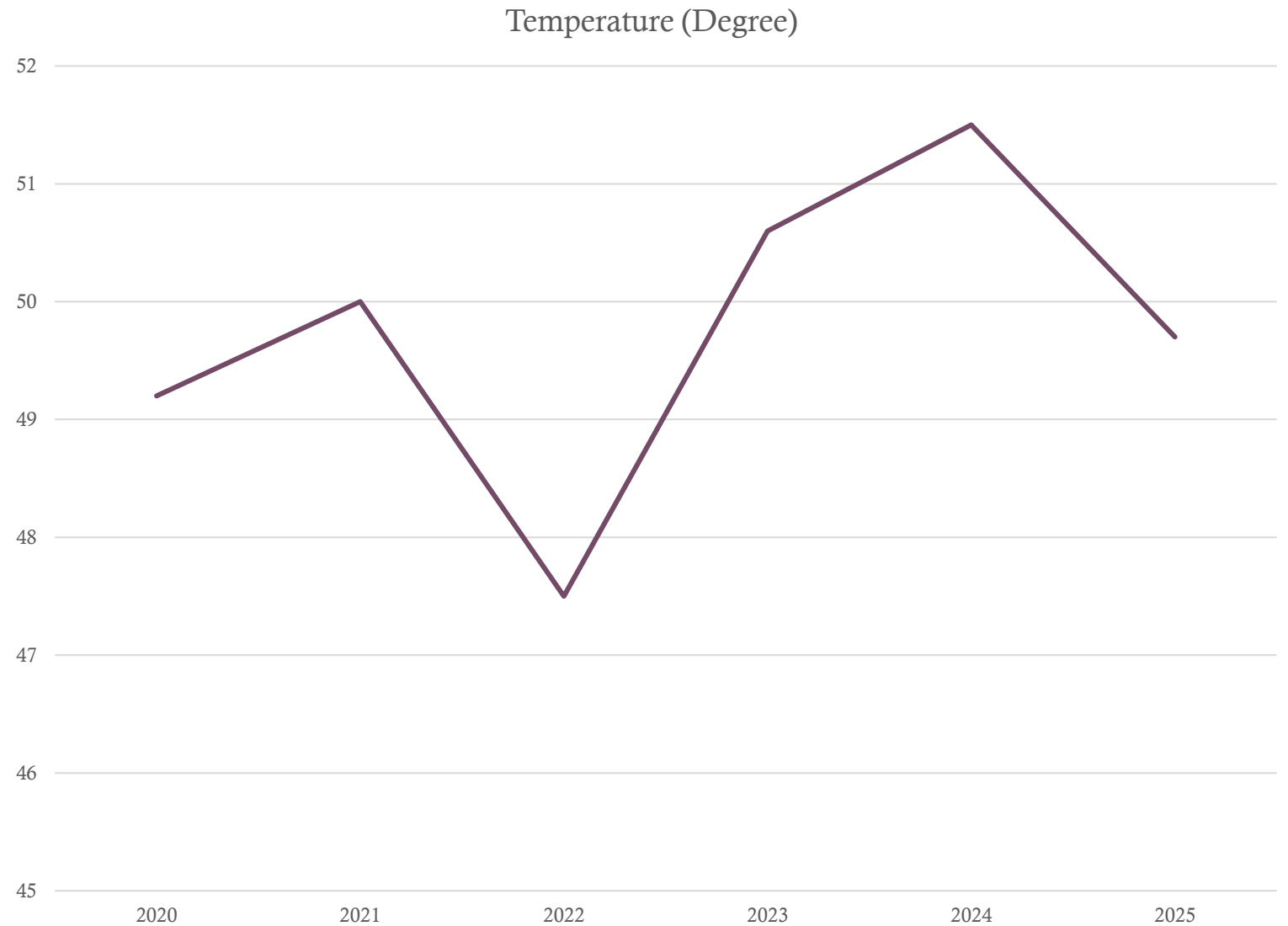
X_3 (NITROGEN FERTILIZER):
TOTAL POUNDS APPLIED PER
ACRE IN IOWA



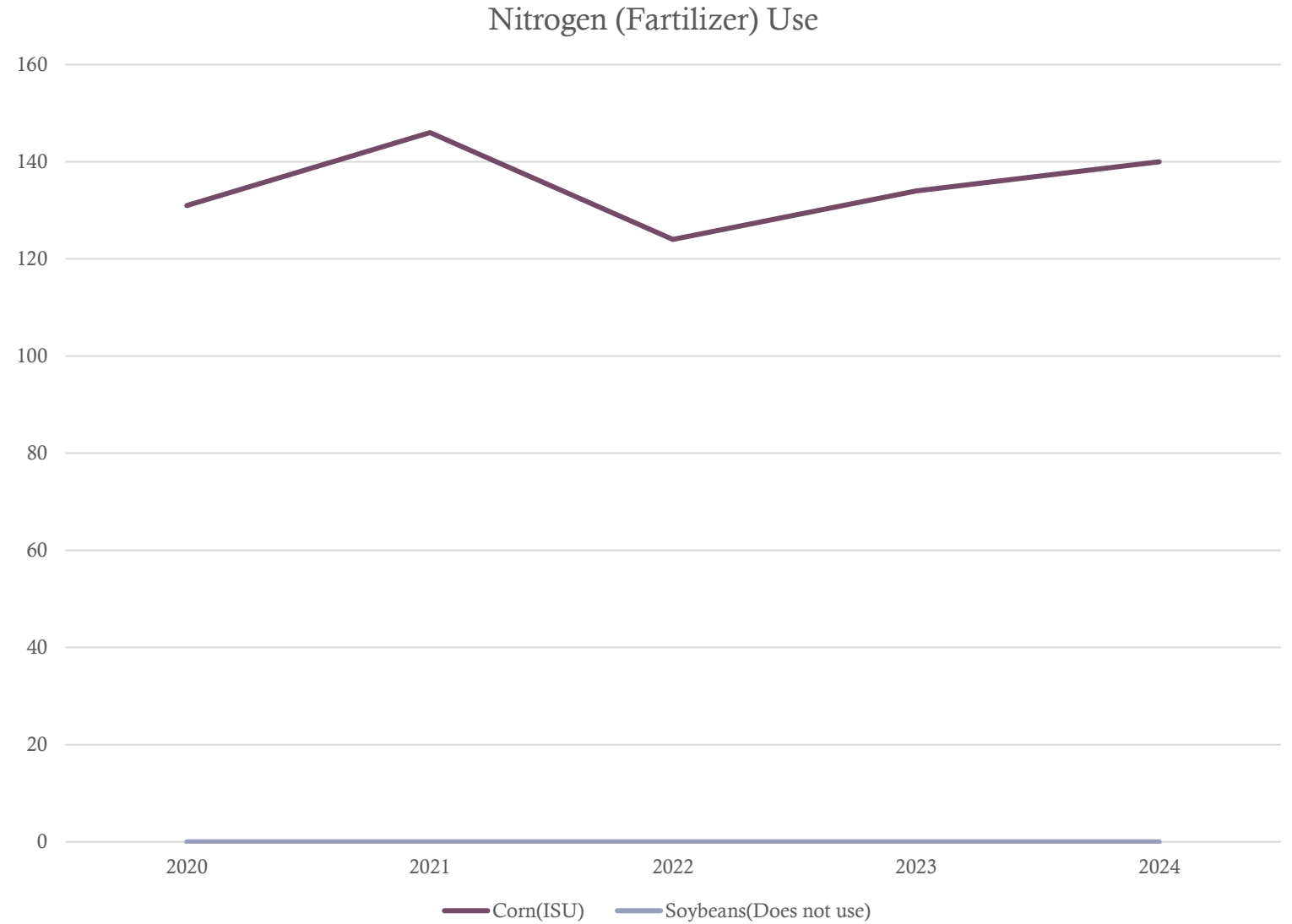
PRECIPITAION DATA IN RELATION TO CORN & SOYBEANS



TEMPERATURE DATA IN RELATION CORN & SOYBEANS



NITROGEN (FERTILIZER) USE FOR CORN & SOYBEANS



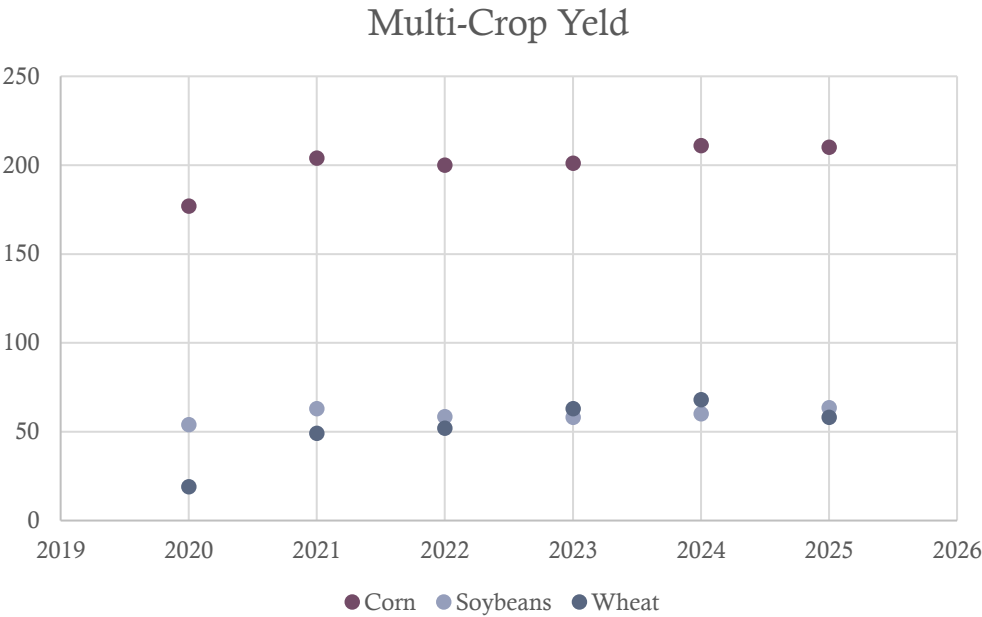
QUICK NOTE

- Due to missing data
 - Wheat yield data for the Iowa in the following slides was estimated using regional Soft Red Winter
 - Wheat averages from USDA NASS Small Grains Summaries, as state-level data for Iowa is
 - frequently suppressed due to low harvested acreage.
- The "Regional Representative" Proxy: The Soft Red Winter Wheat (SRW) average for the "Corn Belt" region.

THE OUTCOME(MULTI-CROP YIELD) MATRIX (X)

2020-2025

Corn(Y_C)	Soybean(Y_S)	Wheat(Y_W)
54	231	19
63	267	49
58.5	258.5	52
58	259	63
60	271	68
63.5	273.5	58



PREDICTOR MATRIX (ENVIRONMENTAL PREDICTORS) (X_1, X_2, X_3) 2020-2025

Precipitation (X_P)	Temperature (X_T)	Nitrogen (X_N)	
49.2	28.92	131	
50	31.06	146	
47.5	27.25	124	
50.6	26.82	134	
51.5	36.95	140	
49.7	33.59	172	6x3

Due to Soybeans not requiring nitrogen(therefore=0) and the lack of data for wheat (N/A) within that time period, we will use only the nitrogen data for corn.

COMPUTING Y: THE REVENUE

2020-2025

Corn(Y_C)	Soybean(Y_S)	Wheat(Y_W)	Prices				Revenue
54	231	19	\$3.81	\$11.1	\$5.05	\$3.81	1369.72
63	267	49	\$5.9	\$13.1	\$7.63	\$5.9	2402.77
58.5	258.5	52	\$6.4	\$14.1	\$8.83	\$6.4	2564.01
58	259	63	\$5.5	\$12	\$6.96	\$5.5	2217.46
60	271	68	\$5.5	\$10	\$6.72	\$5.5	
63.5	273.5	58	\$4.3	\$10.5	\$5.1	\$4.3	

$Y =$

Every Row of Y is the dot product of each of the rows of the multi-crop yield matrix and the price matrix (not in the usual matrix multiplication) respectively.

FINDING B (COEFFICIENTS)

$$B=(X^T X)^{-1} X^T Y$$

Multiplied

$$X^T = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 49.2 & 50 & 47.5 & 50.6 & 51.5 & 49.7 \\ 28.92 & 31.06 & 27.25 & 26.82 & 36.95 & 33.59 \\ 131 & 146 & 124 & 134 & 140 & 172 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & 49.2 & 28.92 & 131 \\ 1 & 50 & 31.06 & 146 \\ 1 & 47.5 & 27.25 & 124 \\ 1 & 50.6 & 26.82 & 134 \\ 1 & 51.5 & 36.95 & 140 \\ 1 & 49.7 & 33.59 & 172 \end{pmatrix}$$

$$(X^T X) = \begin{pmatrix} 6 & 298.5 & 184.59 & 847 \\ 298.5 & 14859.59 & 9199.679 & 42174 \\ 184.59 & 9199.679 & 5756.556 & 26246.64 \\ 847 & 42174 & 26246.64 & 120993 \end{pmatrix}$$

$$(X^T X)^{-1} = \begin{pmatrix} 345.8576 & -7.55298 & 1.43352 & -0.0994 \\ -7.55298 & 0.17343 & -0.03818 & 0.000705 \\ 1.43352 & -0.03818 & 0.027396 & -0.00267 \\ -0.0994 & 0.000705 & -0.00267 & 0.001038 \end{pmatrix}$$

FINDING B (COEFFICIENTS) CONT'D

$$B = (X^T X)^{-1} X^T Y$$

Multiply

$$(X^T X)^{-1} = \begin{pmatrix} 345.8576 & -7.55298 & 1.43352 & -0.0994 \\ -7.55298 & 0.17343 & -0.03818 & 0.000705 \\ 1.43352 & -0.03818 & 0.027396 & -0.00267 \\ -0.0994 & 0.000705 & -0.00267 & 0.001038 \end{pmatrix}$$
$$X^T = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 49.2 & 50 & 47.5 & 50.6 & 51.5 & 49.7 \\ 28.92 & 31.06 & 27.25 & 26.82 & 36.95 & 33.59 \\ 131 & 146 & 124 & 134 & 140 & 172 \end{pmatrix}$$
$$(X^T X)^{-1} X^T = \begin{pmatrix} 2.68655 & -1.77913 & 13.82846 & -11.1962 \\ -0.03211 & 0.0355 & -0.26811 & 0.292985 \\ -0.0024 & -0.01437 & 0.035443 & -0.12139 \\ -0.00601 & 0.004404 & -0.01001 & 0.003695 \end{pmatrix}$$

FINDING B (COEFFICIENTS) CONT'D

$$B = (X^T X)^{-1} X^T Y$$

$$(X^T X)^{-1} X^T = \begin{pmatrix} 2.68655 & -1.77913 & 13.82846 & -11.1962 \\ -0.03211 & 0.0355 & -0.26811 & 0.292985 \\ -0.0024 & -0.01437 & 0.035443 & -0.12139 \\ -0.00601 & 0.004404 & -0.01001 & 0.003695 \end{pmatrix}$$

Multiply

$$Y = \begin{pmatrix} 1369.72 \\ 2402.77 \\ 2564.01 \\ 2239.98 \\ 2217.46 \\ 1865.55 \end{pmatrix} \quad (X^T X)^{-1} X^T Y = \begin{pmatrix} 3612.275 \\ -19.1548 \\ 9.985724 \\ -6.06813 \end{pmatrix}$$

THE COEFFICIENT B

$$B=3612.275-19.1548X_1+9.985724X_2-6.06813X_3$$



Point of Intersection

FINDING E THE ERROR

- $E = Y - \hat{Y}$
 - **Y**: Actual
 - $\hat{Y}$: Predicted

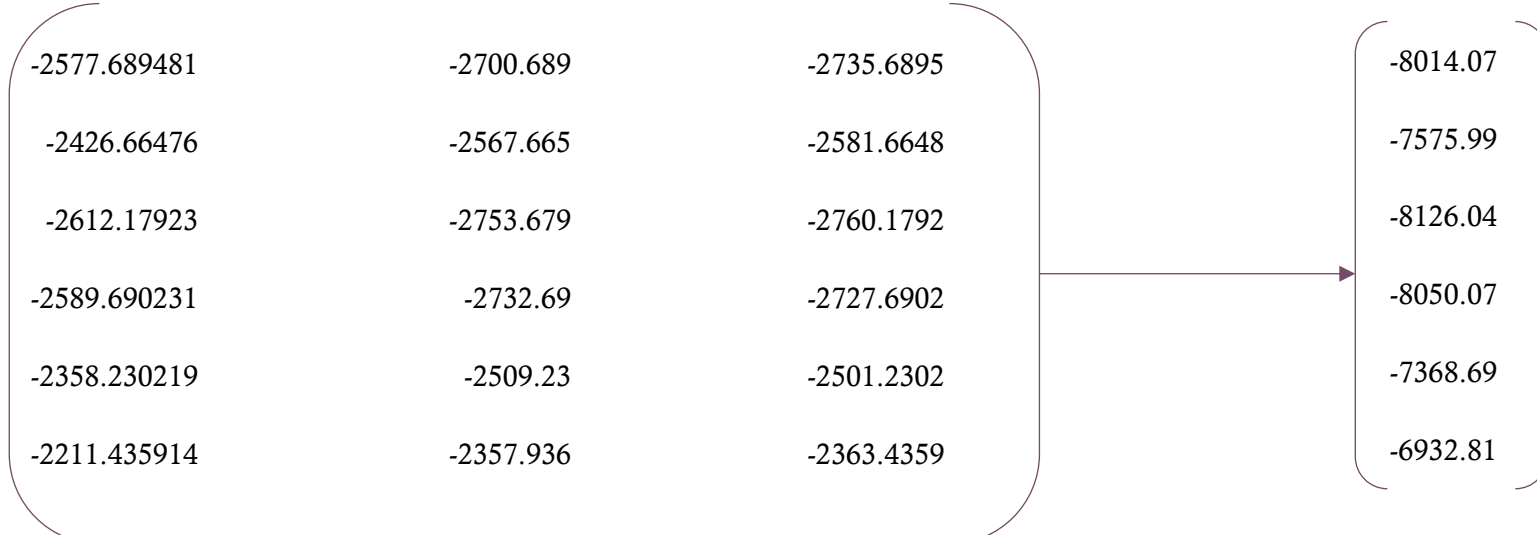
Corn			Soybeans			Wheat		
Y	$\hat{Y}$	E	Y	$\hat{Y}$	E	Y	$\hat{Y}$	E
54	2754.6895	-2577.689481	231	2754.6895	-2700.689	19	2754.6895	-2735.6895
63	2630.6648	-2426.66476	267	2630.6648	-2567.665	49	2630.6648	-2581.6648
58.5	2812.1792	-2612.17923	258.5	2812.1792	-2753.679	52	2812.1792	-2760.1792
58	2790.6902	-2589.690231	259	2790.6902	-2732.69	63	2790.6902	-2727.6902
60	2569.2302	-2358.230219	271	2569.2302	-2509.23	68	2569.2302	-2501.2302
63.5	2421.4359	-2211.435914	273.5	2421.4359	-2357.936	58	2421.4359	-2363.4359

ERROR MATRIX

$E =$

-2577.689481	-2700.689	-2735.6895	-8014.07
-2426.66476	-2567.665	-2581.6648	-7575.99
-2612.17923	-2753.679	-2760.1792	-8126.04
-2589.690231	-2732.69	-2727.6902	-8050.07
-2358.230219	-2509.23	-2501.2302	-7368.69
-2211.435914	-2357.936	-2363.4359	-6932.81

Error Sumed per year



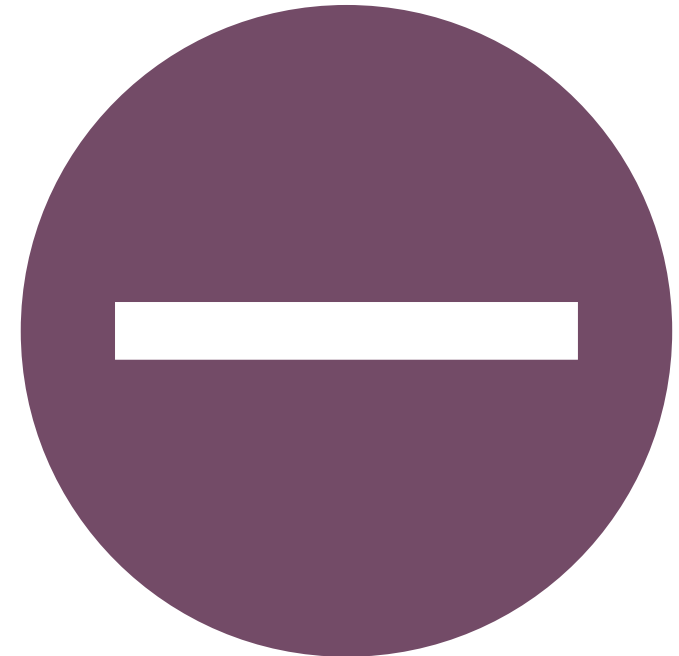
OUR MODEL

$$Y = XB + E$$

$$\begin{pmatrix} 1369.72 \\ 2402.77 \\ 2564.01 \\ 2239.98 \\ 2217.46 \\ 1865.55 \end{pmatrix} = \begin{pmatrix} 1 & 49.2 & 28.92 & 131 \\ 1 & 50 & 31.06 & 146 \\ 1 & 47.5 & 27.25 & 124 \\ 1 & 50.6 & 26.82 & 134 \\ 1 & 51.5 & 36.95 & 140 \\ 1 & 49.7 & 33.59 & 172 \end{pmatrix} \begin{pmatrix} 3612.275 \\ -19.1548 \\ 9.985724 \\ -6.06813 \end{pmatrix} + \begin{pmatrix} -8014.07 \\ -7575.99 \\ -8126.04 \\ -8050.07 \\ -7368.69 \\ -6932.81 \end{pmatrix}$$

ERROR/RESIDUAL COVARIANCE ANALYSIS

- Conclusion of Error
 - We see that the error is intense (all negatives), the predicted far exceeded the actual(yield)
 - This can lead us to conclude that, the three variables or predictors (precipitation, temperature, and nitrogen) should in theory positively influence yield.
 - Something else might be at play



CONCLUSION:

THE REALITY OF THE MATTER

THE LURKING (CONFOUNDING) VARIABLES

- As mentioned previously, the survey reveals that there are financial and economical reasons that affects the yield of these crops, these are the lurking or confounding variables.
 - They caused distortion in the data.
 - They caused false correlations
 - The environmental factors alone don't determine the yield of crops.



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QUESTIONS?



THE END

